



A Simulation-Based Framework for Evaluation of Healthcare Systems with Interacting Factors and Correlated Performance Measures

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Abstract

All multi-criteria decision making (MCDM) techniques used with discrete event simulation (DES) in previous literature in the analysis of healthcare systems assume that performance measures are independent, while in fact many are correlated. In this study, correlations between performance measures are taken into consideration using modified TOPSIS. A framework is developed in which DES is used to model multiple experiments and factor levels to estimate various performance measures. Design of experiments is used to study the effects of factors and their interactions. Experiments are input to modified TOPSIS as alternatives to be compared based on multiple correlated performance measures. A case study is presented at Educational Dentistry Clinics in Jordan. This study was designed to improve overall clinic performance. Results show that when using appropriate appointment scheduling rules and orthodontist working schedules, and exploiting the resources effectively, the best scenario obtained shows an effective decrease in orthodontic patient waiting times and length of stay (LOS), and primary diagnosis patients LOS by 29.44%, 37.74% and 7.12%, respectively. It also shows an increase in orthodontist and receptionist utilizations, and number of patients served by 11.22%, 65.82% and 69.39%, respectively. The results obtained from modified TOPSIS were more accurate as compared to traditional TOPSIS. As a future work, other appointment rules can be tested and other MCDM techniques that take criteria correlations into consideration can be used to rank the experiments such as the analytic network process.

Keywords Discrete event simulation · Healthcare systems · Dental Clinics · Design of experiments · Modified TOPSIS

1 Introduction

The increasing demand for healthcare services in hospitals and healthcare centers has led to overcrowding of patients in these sectors, because of pressure on limited capacity resources and the disproportionate increase in the number of patients. Patients end up waiting for long periods of time to receive a service, this waiting may have a negative impact on patients' health and thus their satisfaction about the service

provided. Thus, there is great importance in improving the efficiency of healthcare systems by organizing patient appointments, utilizing resources effectively, etc.

Improvement of healthcare systems without using an appropriate tool becomes a tedious and inefficient task. Simulation is an effective tool that can deal with healthcare systems challenges [1]. In the last 50 years, the number of publications on simulation in healthcare has increased significantly because of its effectiveness in improving systems efficiency [2].

Discrete event simulation (DES) is the main simulation tool that is used widely in healthcare applications. It is a computer-simulation modeling tool that provides a systematic approach to imitate real systems [3]. The most important advantages of DES are its ability to implement “what if” scenarios analyses in a constructed simulation model and find the best solutions that improve system performance before being implemented in the real world [4]. The performance measures, such as doctor and nurse utilizations, patient waiting times and length of stay

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and number of patients served, can be examined and compared for each suggested scenario [5].

In this paper, a case study is implemented in the Educational Dentistry Clinic (EDC) located in Jordan. It is one of the most important dental clinics in the region. The clinic includes a number of specialist dentists and is provided with the latest medical devices to provide optimal dental services at local and global levels. It consists of five floors; the base floor is for radiography imaging which contains: panoramic, lateral cephalometric, computed tomography (C.T) scan, and traditional periapical radiography. The ground floor contains one emergency department and seven private clinics which are: orthodontic, primary diagnosis, periodontics, restorative, surgery, kids and prosthodontics in addition to digital periapical radiography, one reception room and one accounting room. The next three floors are for training dentistry students. This research focuses on base and ground floors, and the other floors were not considered. All clinics share the same registration, accounting and radiography rooms. Each clinic has its own dentists and each dentist has a specific schedule and his/her own patients.

The clinic's working hours are from 8:00 a.m. to 4:00 p.m., five days a week (Sunday–Thursday). All clinics except emergency receive patients by appointments which are organized by receptionists. Some patients might miss their appointments or come to the clinic without an appointment which are called no-shows and walk-ins, respectively, and are dealt with in accordance to specific rules.

This paper focuses on orthodontic and primary diagnosis clinics only. The remaining clinics were constructed in the simulation model so that their effect on shared resources such as reception, accounting and radiography rooms is taken into consideration.

The aim of this research is to develop a framework for analyzing and optimizing EDC in which DES is used to model multiple scenarios and factor levels to estimate various performance measures. DOE is used to study the effects of factors and their interactions. Each combination of factor levels is then input to modified TOPSIS technique as alternatives to be compared based on multiple correlated criteria represented by the performance measures with the goal of selecting the best alternative. Performance measures include patients' waiting times, length of stay (LOS), throughput and resource utilizations. Factors considered are resource allocation, patient flow scenarios and patient appointment schedules.

2 Literature Review

2.1 Effects of Patient Flows, Appointment Scheduling and Capacity Allocation on Performance Measures

Many researchers studied the impact of patient flows, capacity management and patient appointment schedules separately on selected performance measures; for example [6], tested several appointment rules to improve the performance of computed tomography scan facility, and the study considered different patients categories such as outpatients, inpatients, emergency arrivals and walk-ins. Three scenarios were suggested using DES in an outpatient internal medicine department to study the effect of patient flows on patient waiting times [7]. Results showed significant improvements in waiting times, which were reduced by 71.28%, and [8] used DES in an internal medicine outpatient clinic to study the effect of resource allocation, patient priorities and patient volumes on resource utilization, patient waiting times and number of patients served. Several scenarios were tested, and the results showed a reduction in patient waiting times by 20%, increase in number of patients served by 54.8% and an increase in utilization by 27.5 % when patients had no priority. On the other hand, several researchers studied the combination of the factors and their effects on the overall system performance. DES was used to improve the performance in an outpatient orthopedic clinic [9]. The authors suggested three factors to test different scenarios, the factors were patient appointment schedules, added an extra X-ray technician and ensured medical staff arrived when the clinic opened. Different scenarios were modeled based on the combination of these factors while considering several performance measures. DES was used in an ambulatory care unit to study the impact of resource allocation, patient appointment schedules, the time that the clinic opened and number of medical students on selected performance measures such as resource utilizations, waiting times and clinic overtime under different scenarios [10]. Few researchers used design of experiments (DOE) to study the impact of different factors on selected performance measures. DES was integrated with DOE to study the interactions between consulting rooms allocation, patient appointment schedules and patient flows through radiology at different levels [11]. They investigated how these factors jointly affect patient waiting times, utilization of resources and clinic overtime. The interactions between appointment scheduling, resource allocation and patient trajectories in an outpatient orthopedic clinic on patient length of stay, number of patients seen by each orthopedist and proportion of occupied time for each orthopedist were studied



using DES and DOE [12]. DES was used with DOE to test the impact of different appointment schedules, rearranging new and old patient slots and adding a new imaging technology on patient waiting times [13]. In this research, DES and DOE are integrated to study the interaction effect of patient appointment schedules, patient trajectories and resource allocations on patient LOS, waiting times as well as dentist and receptionist utilizations.

2.2 Integrated Multi-Criteria Decision Making (MCDM) with DES

The purpose of using MCDM is to rank suggested scenarios in a model based on generated performance measures. A few number of articles integrated DES with MCDM techniques in healthcare systems analysis. For example, DES was integrated with analytic hierarchy process (AHP) and technique for order of preference by similarity to ideal solution (TOPSIS) to rank fourteen suggested scenarios [14]. Also, DES was used with AHP to select the best scenario that reduced patient waiting times, increased patient satisfaction and improved quality of patients care [15]. DES with interval type-2 FAHP and ELECTRE decision tools was used by [16] to select the best scenario that improved the performance of an emergency department by reducing patient lead times. The study used interval type-2 FAHP to weigh and select performance measures such as LOS, doctor utilizations and number of patients served, and then, ELECTRE decision tool was used to rank and select the best scenario among five. In order to avoid difficulties in the ELECTRE method, fuzzy PROMETHEE decision tool was used [17]. The researchers study the effect of adding general or specialist physicians in an emergency department on the number of patients served and patient LOS.

All MCDM techniques used with DES in previous literature assume that the performance measures are independent, while in fact many are dependent and correlated. In this study, the correlation between performance measures will be studied using the modified TOPSIS technique. The traditional TOPSIS assumes that all criteria are independent [18]. Euclidean distances do not take into consideration the correlation between criteria. In contrast, Mahalanobis distances [19] were developed to take variation as well as correlation between performance measures into consideration. The competitiveness of Chinese high-tech industry was evaluated using modified TOPSIS with Mahalanobis distances instead of Euclidean distances to improve their results [20]. The indicators used were correlated, and the results became more accurate with modified TOPSIS compared to the traditional one.

Up to our knowledge, this is the first study that uses DES and DOE with modified TOPSIS which takes into

consideration correlations between performance measures to select the best combinations of factors in dental clinics.

3 Research Methodology

3.1 Traditional TOPSIS

This technique was developed by Hwang and Yoon in 1981 which is a MCDM tool. It is used to select the best alternative among many that has the shortest Euclidean distance from ideal solution and the farthest Euclidean distance from negative ideal solution based on multiple independent criteria. Equation (1) shows a general format for a decision matrix used in traditional TOPSIS. It has a set of m alternatives ($A_i, i = 1, 2, \dots, m$), a set of n criteria ($C_j, j = 1, 2, \dots, n$) and a set of n subjective criteria weights determined by experts ($w_j, j = 1, 2, \dots, n$), where ($X_{ij}, i = 1, 2, \dots, m, j = 1, 2, \dots, n$) represent criteria values for specific alternatives. Prior to TOPSIS analysis alternatives must be chosen according to various combinations of factors. Criteria on the other hand represent performance measures used as a basis for comparison which must also be decided. DES models are then built for each alternative and run to obtain their specific performance measures or criteria values. Finally, TOPSIS analysis is performed to select the best alternative based on multiple criteria.

$$\begin{pmatrix} \text{Alternatives} & C_1 & C_2 & \cdots & C_n \\ A_1 & X_{11} & X_{12} & \cdots & X_{1n} \\ A_2 & X_{21} & X_{22} & \cdots & X_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ A_m & X_{m1} & X_{m3} & \cdots & X_{mn} \\ \text{Weight} & w_1 & w_2 & \cdots & w_n \end{pmatrix} \quad (1)$$

The steps of how traditional TOPSIS is performed are as follows:

Step 1: The decision matrix is normalized using Equation (2):

$$r_{ij} = \frac{X_{ij}}{\sqrt{\sum_{i=1}^m X_{ij}^2}}, i = 1, 2, \dots, m; j = 1, 2, \dots, n \quad (2)$$

Step 2: The weighted normalized decision matrix is calculated as shown in Equation (3):

$$v_{ij} = w_j r_{ij}, i = 1, 2, \dots, m; j = 1, 2, \dots, n \quad (3)$$

Step 3: The ideal positive solution (S^+) and ideal negative solution (S^-) are calculated as shown in Equations (4) and (5).



$$S^+ = (s_1^+, s_2^+, \dots, s_n^+) \quad (4)$$

$$S^- = (s_1^-, s_2^-, \dots, s_n^-) \quad (5)$$

For benefit criteria (the more, the better):

$$s_j^+ = \max \{v_{ij} | 1 \leq i \leq m\} \text{ and } s_j^- = \min \{v_{ij} | 1 \leq i \leq m\}.$$

For cost criteria (the less, the better):

$$s_j^+ = \min \{v_{ij} | 1 \leq i \leq m\} \text{ and } s_j^- = \max \{v_{ij} | 1 \leq i \leq m\}.$$

Step 4: For each alternative, the Euclidean distance from ideal positive solution and ideal negative solution is calculated as shown in Equations (6) and (7).

The distance between the alternative A_i and the ideal positive solution is:

$$d_i^+ = \sqrt{\sum_{j=1}^n (s_j^+ - v_{ij})^2}, i = 1, 2, \dots, m \quad (6)$$

The distance between the alternative A_i and the ideal negative solution is:

$$d_i^- = \sqrt{\sum_{j=1}^n (s_j^- - v_{ij})^2}, i = 1, 2, \dots, m \quad (7)$$

Step 5: The score for each alternative is calculated using Equation (8):

$$R_i = \frac{d_i^-}{d_i^- + d_i^+}, i = 1, 2, \dots, m \quad (8)$$

Finally, the alternatives are ranked based on their scores, and the alternative with the highest score is the best choice.

3.2 Modified TOPSIS

3.2.1 Mahalanobis Distance

Euclidean distances do not take into consideration correlations between criteria. In contrast, Mahalanobis distances take correlations between criteria into consideration. The Mahalanobis distance between two vectors from the same dataset is shown in Eq. (9) as follows:

$$d(\vec{x}, \vec{y}) = \sqrt{(\vec{x} - \vec{y})^T S^{-1} (\vec{x} - \vec{y})} \quad (9)$$

where $(\vec{x} - \vec{y})^T$ is transpose of the difference between the two vectors. S^{-1} is covariance matrix inverse.

It should be noted that if criteria are independent and no correlations exist the covariance matrix is replaced by an

identity matrix in the Mahalanobis distance and is reduced to a Euclidean distance. On the other hand, if the covariance matrix is diagonal, the Mahalanobis distance is reduced to a normalized Euclidean distance.

Moreover, weights can be used to distinguish the relative importance of variables (in this study the criteria or performance measures), Eq. (10) represents the weighted Mahalanobis distance [21]:

$$d(\vec{x}, \vec{y}) = \sqrt{[(\vec{x} - \vec{y})\Omega]^T S^{-1} [(\vec{x} - \vec{y})\Omega]} = \sqrt{(\vec{x} - \vec{y})^T \Omega^T S^{-1} (\vec{x} - \vec{y})\Omega} \quad (10)$$

where Ω is a diagonal matrix of criteria weights made up from criteria weights present in the decision matrix.

$$\Omega = \text{diag}(w_1, w_2, \dots, w_n) \quad (11)$$

3.2.2 Modified TOPSIS

Mahalanobis distance is used instead of Euclidean in modified TOPSIS. The steps of applying modified TOPSIS are as follows:

Step 1: The decision matrix is normalized similar to step one in traditional TOPSIS, Eq. (2) is applied, the second step in traditional TOPSIS is neglected in modified TOPSIS, and the weight of each criterion is already introduced to Mahalanobis distance as shown in Eq. (10)

Step 2: The ideal positive and the ideal negative solutions are computed similar to step three in traditional TOPSIS; Eqs. (4) and (5) are applied.

Step 3: For each alternative, Mahalanobis distance is calculated from ideal positive solution and ideal negative solution as shown in Eqs. (12) and (13).

The distance between an alternative A_i and the ideal positive solution is:

$$d(r_i, S^+) = \sqrt{\left\{s_j^+ - r_{ij}\right\}\Omega S^{-1}\Omega^T \left\{s_j^+ - r_{ij}\right\}^T}, i = 1, 2, \dots, m \quad (12)$$

The distance between an alternative A_i and the ideal negative solution is:

$$d(r_i, S^-) = \sqrt{\left\{s_j^- - r_{ij}\right\}\Omega S^{-1}\Omega^T \left\{s_j^- - r_{ij}\right\}^T}, i = 1, 2, \dots, m \quad (13)$$

Step 4: The score of each alternative is calculated using Eq. (14):

$$R_i = \frac{d(r_i, S^-)}{d(r_i, S^-) + d(r_i, S^+)}, i = 1, 2, \dots, m \quad (14)$$



The alternatives are ranked based on their scores, and the alternative with the highest score is the best choice.

3.3 Discrete Event Simulation

Patient flows from entrance to departure and the interactions of each staff member with patients are studied. For this reason, interviews with medical and non-medical staff in addition to direct observations of patient flows during the study period have been performed. Four types of patients arrive to the dental clinics: Type 1: patients enter to inquire about dentist schedules or types of clinics available, and Type 2: patients enter to make an appointment, these two types do not require a dentist, and they only pass through reception and then depart. Type 3: follow-up patients, who have existing records in the system; they go through registration to confirm appointments for a specific clinic. Type 4: New patients, who need new records to be opened and then go through registration to confirm appointments for a specific clinic. Sometimes a dentist can see patients without an appointment which are called walk-ins. Types 3 and 4 patients then go to dentist examinations. A brief description of how follow-ups and new patients arrive and move through the EDC is shown in Fig. 1.

The clinics have two receptionists, two radiographers in each floor, one accountant and a number of dentists in each clinic with different schedules. Orthodontic and primary diagnosis clinics have three dentists, two rooms and three dentists, one room, respectively.

For the orthodontic clinic, the clinic deals with patients from the previous day that come to put orthodontic cases (this operation needs more service time compared to other orthodontist operations), and the remaining patients are either new or follow-ups. For new patients, the dentist performs a preliminary medical examination to decide the type of radiography that should be taken, and the radiography processes that the orthodontist needs are either panoramic with lateral cephalometric or C.T scan. The patients then go to the radiography room based on dentist's request. When finished, patients go to accounting to pay for the

radiographs and to receive a receipt. When done, patients go back to the radiographer and give them the receipt, so patients can take the radiographs, or have them sent to the dentist via a computer network. Then, the patients go back to the clinic and the dentist diagnoses their condition to begin the treatment. For follow-ups, the dentist starts treatment directly. After the patients complete their treatment, they go to accounting to pay for dentist's treatment and then either go back to reception to make another appointment or they have finished their treatments completely and depart.

Patient flows for primary diagnosis clinic are similar to the orthodontic clinic except that all dentist operations have the same service times and radiography processes needed are digital periapical and panoramic. The service times for each step of the patient flows in orthodontic (37 observations) and primary diagnosis clinics (55 observations) are obtained using input analyzer of Rockwell Arena. The times are shown in Table 1. Since *P*-value is more than 0.05 for all service times, the data for each process follow the corresponding distribution.

The percentages of new patients in orthodontic and primary diagnosis clinics are 6% and 77%, respectively. The remaining patients are follow-ups.

Inter-arrival times for all patients of all clinics were collected using a time study during a period of 18 days and include all patient walk-ins and no-shows. Figure 2 shows the average number of patients that arrive in each hour per day in a week. The number of patients that arrive varies from day to day, with minimal differences between weeks. On Sunday and Thursday the clinics receive the largest number of patients. In contrast, on Wednesday a few number of patients arrive, this difference is due to the number of clinics available during the week, and most clinics work on Sunday and Thursday. Also, the figure shows a few number of patients arrive during the first and the last hour of the working day. The data were collected directly from EDC in order to construct a DES model that simulates the real system with Rockwell's ARENA software. The base model was built based on real inter-arrival times of patients.

Fig. 1 Patients flowchart in EDC

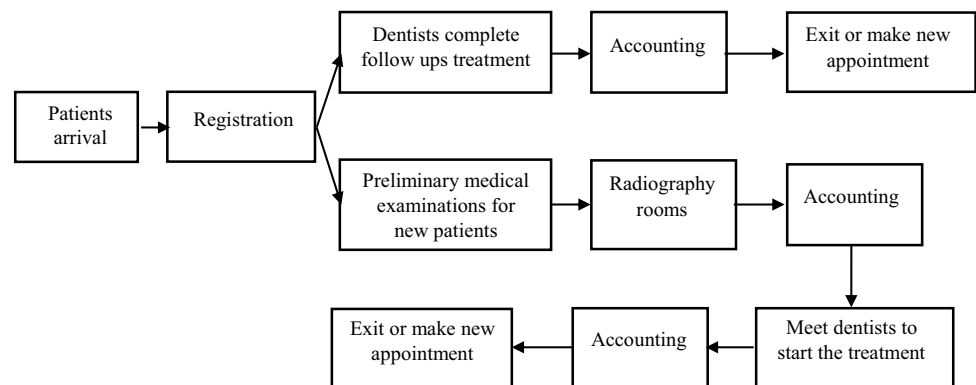
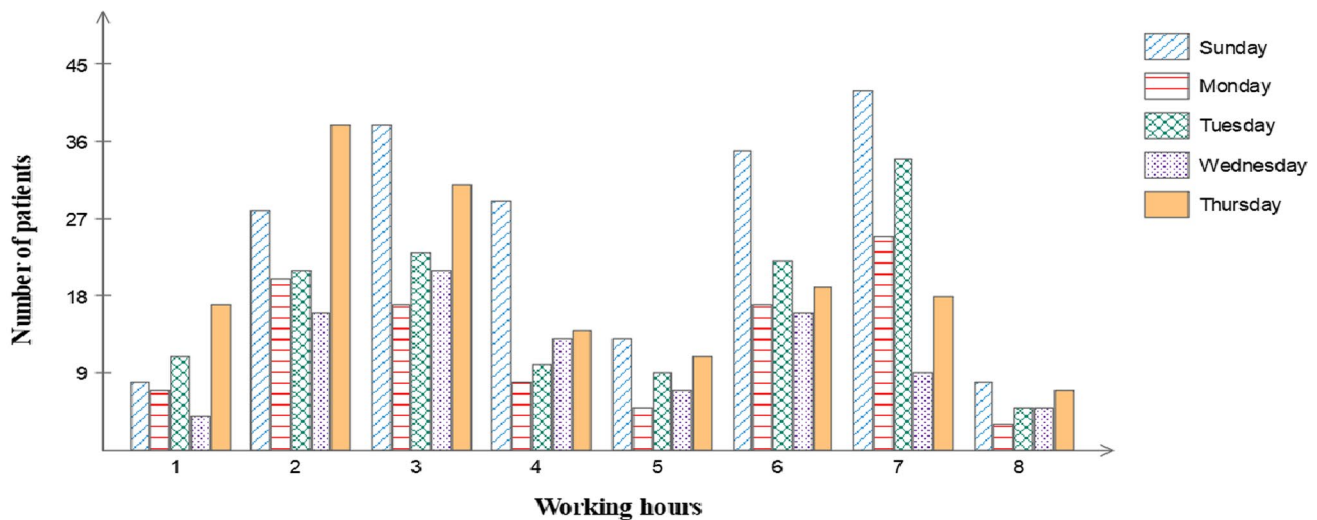


Table 1 Statistical distributions used in simulation model for service times

Process	Mean of service times (min)	Standard deviation (min)	Expression (Min)	P-value
New patient registration	2.06	0.962	$0.18 + \text{LOGN}(1.87, 0.91)$	0.544
Old patient registration	0.464	0.381	$\text{LOGN}(0.46, 0.377)$	0.127
Make an appointment	1.49	0.769	$\text{LOGN}(1.51, 0.847)$	0.200
Ask a question	0.761	0.644	$\text{LOGN}(0.745, 0.564)$	0.192
Accounting process	1.28	0.513	$\text{TRIA}(0.18, 1.49, 2.42)$	0.192
Preliminary medical examination	7.98	7.42	$0.999 + \text{WEIB}(6.55, 0.878)$	0.118
Orthodontist treatment	10.2	6.15	$1 + \text{ERLA}(4.59, 2)$	0.487
Orthodontic cases service	–	–	$\text{Uniform}(20, 30)$	–
Primary diagnosis dentist	–	–	$\text{Uniform}(15, 25)$	–
Panoramic radiography	3.92	0.911	$2.39 + \text{LOGN}(1.58, 1.19)$	> 0.15
lateral cephalometric radiography	3.92	0.911	$2.39 + \text{LOGN}(1.58, 1.19)$	> 0.15
Digital periapical radiography	2.95	1.59	$1 + \text{GAMM}(1.15, 1.69)$	0.541

**Fig. 2** Number of patients that arrive per hour for one week

3.4 Simulation Model Assumptions and Validation

It is difficult to build a simulation model that is completely identical to the real system, so assumptions were used during EDC model construction in order to simplify the simulation model and reduce its complexity. The assumptions are summarized in the following points:

- All radiography processes receive patients in first-come-first-serve (FCFS) manner.
- All new patients sent to take a radiograph in all clinics except orthodontic have a higher priority to see the dentists when they come back over patients who are just arriving.
- Patients who put orthodontic cases have higher priority to see the orthodontist followed by new patients who are coming back from radiography followed by follow-ups.
- New patient's preliminary medical examination service times are the same for all clinics.
- Some processes' service times are estimated by specialized dentists due to their long durations or because of their infrequent occurrence. For this reason, uniform or triangular distributions are used.
- All dentists within the same clinic have the same service time, except for orthodontists and prosthodontics.

Simulation model validation is an approach to compare the results of the constructed simulation model with real



Table 2 Comparison of number of patients served in simulated model and real system

The clinic	Average simulated number of patients served	Real number of patients served	Percentage difference (%)	Half width
Emergency	160.56	157	2.267	< 1.67
Orthodontic	400.99	402	0.251	< 2.18
Primary diagnosis	36.63	38	3.605	< 0.77
Periodontics	149.35	149	0.255	< 1.49
Restorative	85.72	85	0.847	< 1.27
Surgery	123.86	121	2.363	< 1.39
Kids	78.41	80	1.992	< 1.09
Prosthodontics	185.08	185	0.043	< 1.66

system results. In this paper, the number of patients served in each clinic in the model is compared to the number of patients served in each clinic in the real system during the research period of 18 days. So, 200 independent replicates were run to validate the model, and Table 2 shows that the percentage difference of patients served in each clinic is within 5%, so the simulation model is valid and represents the real system.

3.5 Design of Experiments, Decision Factors and Performance Measures

This research work focuses on three places at EDC which are orthodontic clinic, primary diagnosis clinic and reception. Medical and non-medical staff as well as the constructed model confirm significant problems related to these three places. Firstly, orthodontic clinic is the most crowded clinic, the main problems related to this clinic are that patients arrive randomly, the current appointment rule is irregular, and the resources are not well exploited which leads to high orthodontic patients waiting times and LOS compared to their treatment times due to crowdedness. Secondly, all new patients need to visit the radiography department and there are too many unnecessary steps in their flow paths especially primary diagnosis patients. 77% of primary diagnosis patients are new, and they go through the current unnecessary steps in their flow paths, so their LOS is affected adversely. Thirdly, two receptionists work in the clinics and their processing times are short, so their utilizations are low. In this research, several factors such as appointment rules for orthodontic patients, resource allocations of orthodontist schedules and number of receptionists and new patients flow paths are tested and the effect of these factors on performance measures is evaluated.

This paper studies four factors with corresponding levels that are related to the EDC which are (1) appointment scheduling rules for orthodontic clinic, (2) number of receptionists, (3) orthodontist working schedules and (4) patient flows

3.5.1 Appointment Scheduling Rules

Appointment rules represent how patients arrive to the clinic; two appointment rules for the orthodontic clinic are tested in addition to the current rule which is irregular and does not take into consideration patients who want to put orthodontic cases. Although the clinic deals with only two patients who need to put orthodontic cases in the morning hours, receptionists give appointments to other patients in the same time slot. The mean and standard deviation of orthodontist service times are similar; all orthodontists do not take more than 15 minutes for new and follow-ups and 30 minutes for patients who want to put an orthodontic case.

The first rule suggested is individual block/30 minute fixed interval for patients who want to put an orthodontic case and individual block/15 min fixed interval for new and follow-up patients. The first two appointments in each day are for the patients who want to put an orthodontic case, and the remaining appointments are for new and follow-ups.

The second rule suggested is individual block/30 minute fixed interval for patients who want to put an orthodontic case and multiple block/30 fixed interval for new and follow-up patients. Like the first rule, the first two appointments are for patients who want to put an orthodontic case with 30 minutes duration, the remaining appointments are for new and follow-ups, and every 30 minutes three appointments are scheduled.

It should be noted that walk-ins and no-shows are taken into consideration when the appointment rules are applied in the simulation model, and 15% of patients miss their appointments in orthodontic clinic. In contrast, one walk-in patient arrives to the clinic every 30 to 60 minutes. This paper does not take into consideration patients lateness or patients who arrive before their appointments.

3.5.2 Resource Allocation

Two factors used in this research are considered as resource allocation factors: The first factor is related to reception in which the number of receptionists is varied, and this factor



Table 3 Weekly orthodontist schedules of orthodontic clinic

Day	Sunday	Monday	Tuesday	Wednesday	Thursday
Morning shift (8:00 a.m.–12:15 p.m.)					
Room 1	Orthodontist 2	Orthodontist 1	–	–	Orthodontist 1
Room 2	–	–	Orthodontist 3	–	–
Afternoon shift (1:00 a.m.–4:00 p.m.)					
Room 1	Orthodontist 2	Orthodontist 1	–	–	Orthodontist 1
Room 2	–	Orthodontist 3	Orthodontist 3	–	–

has two levels: (1) current situation with two receptionists and (2) one receptionist.

The second factor is related to orthodontists work schedules. Table 3 shows weekly orthodontist schedules of the orthodontic clinic. Two orthodontists work schedules are suggested, and they are based on the exploitation of available resources. The orthodontic clinic is closed on Wednesdays, and therefore, new schedules exploit this day. The first schedule suggested is adding one working day in orthodontists schedules which is Wednesday and is specified for patients who want to put an orthodontic case only. The remaining days are for new and follow-ups only. The second schedule also added Wednesdays to the orthodontists working schedules, but all days are the same; the first two appointments are for orthodontic cases, and the remaining appointments are for new and follow-ups.

3.5.3 Patient Flows

Patient flows in the clinics have unnecessary paths that can be eliminated. In the current situation, new patients visit accounting and radiography rooms twice, and the alternative is to have them pay for radiographs and dentist treatments once before departure and have them pick up their radiographs directly when they are completed.

3.5.4 Performance Measures

In this research, the impact of the suggested factors on several clinic performance measures is studied. Each selected performance measure is explained in the following points:

- *Length of stay (LOS)*: is defined as the total time that a patient spends in the clinics. In this research, the average LOS is measured for orthodontic and primary diagnosis patients from their arrival to their departure.
- *Resource utilization*: is defined as the proportion of time that the resource is busy. In this research, average utilizations of orthodontists and receptionists are measured.
- *Waiting time*: is defined as the waiting time for patients before meeting a dentist. The average waiting times of orthodontic patients are measured in this research.
- *Number of patients served*: is the number of patients served from a clinic. The number of patients served from the orthodontic clinic is measured.

3.5.5 Design of Experiments

Thirty-six (36) experiments were performed in the simulation model, each experiment was replicated 20 times, and the total number of experiments is 720. Table 4 summarizes all factors and their corresponding levels; the design

Table 4 Factors and their corresponding levels

Factor	Number of levels	Description of levels
Appointment rule for orthodontic clinic	3	Rule 0: Current rule (no specific rule) Rule 1: One appointment every 15 min Rule 2: Three appointments every 30 min In addition to one appointment every 30 min for the first two appointments
Orthodontists working schedules	3	Schedule 0: Current schedule (the clinic on Wednesday is not available) Schedule 1: Adding Wednesdays to orthodontists working schedules and is specified for patients who want to put orthodontic cases Schedule 2: Adding Wednesdays to orthodontists working schedules but the clinic receives all patient types in all days
Patients flow	2	Flow 0: Current flow, new patients visit accounting and radiography rooms twice. Flow 1: New patients visit radiography and accounting once before they exit from the clinic
Number of receptionists	2	One receptionist Two receptionists



of experiment analysis was performed using Design Expert 11 software.

4 Results and Discussion

4.1 Results of Experimental Design

In this section, the impact of decision factors on the selected performance measures is studied using a full-factorial ANOVA with 5% significant level and the results of each performance measure are analyzed. The following sections contain ANOVA and experimental results on each selected performance measure.

4.1.1 Results of Orthodontic Patients LOS and Waiting Times

Table 5 represents ANOVA on orthodontic patients average LOS and waiting times. As shown, both performance measures showed similar behaviors, and the appointment scheduling rules and orthodontist working schedules have significant effect on both performance measures. (Their *P*-values are less than 0.05.) The appointment rules factor has the highest effect on patients average LOS and waiting times (its mean square values are 63330.66 and 47831.09, respectively), and it demonstrates how necessary it is to choose appropriate appointment rules for patients. In addition, orthodontist working schedules have a higher effect on patients average waiting times than on patients average LOS. (Their mean square values are 1915.47 and 128.24, respectively.) The number of receptionists is not statistically significant, and this means that reducing the number of receptionists to one does not affect patients LOS or waiting times.

In addition, patient flows do not affect either performance measure. The reason is because the percentage of new orthodontic patients that need to take radiographs is only 6% of all orthodontic patients that enter the clinic, and therefore, the number of patients benefiting from changing the flow is low.

ANOVA in Table 5 presents other significant interactions on patients average LOS and waiting times, and the appointment scheduling rules and orthodontist working schedules interaction has significant effect on both measures. Also, patient flows, appointment scheduling rules and orthodontist working schedules interaction has a significant effect on both performance measures.

Patients average LOS and waiting times for all possible combinations of factors (36 experiments) that were performed in simulation models are presented in Fig. 3. Each figure contains 36 points which represent the value of orthodontic patients average LOS and waiting times in each experiment with each experiment being replicated 20 times. The two curves have the same pattern, and they increase and

Table 5 ANOVA on orthodontic patients average LOS and waiting times

Source	Sum of Squares	df	Mean Square	<i>F</i> -value	<i>P</i> -value
<i>Orthodontic patients average LOS</i>					
Model	1.419E+05	35	4054.99	119.18	< 0.0001
A-Receptionists	49.31	1	49.31	1.45	0.2291
B-Patient Flow	46.32	1	46.32	1.36	0.2437
C-Rules	1.267E+05	2	63330.66	1861.37	< 0.0001
D-Schedules	256.48	2	128.24	3.77	0.0236
AB	0.1714	1	0.1714	0.0050	0.9434
AC	109.42	2	54.71	1.61	0.2011
AD	47.18	2	23.59	0.6933	0.5003
BC	21.86	2	10.93	0.3212	0.7254
BD	14.24	2	7.12	0.2092	0.8113
CD	13883.59	4	3470.90	102.01	< 0.0001
ABC	24.79	2	12.39	0.3642	0.6949
ABD	166.04	2	83.02	2.44	0.0879
ACD	125.24	4	31.31	0.9203	0.4515
BCD	421.31	4	105.33	3.10	0.0153
ABCD	97.34	4	24.33	0.7152	0.5817
Pure Error	23272.14	684	34.02		
Cor Total	1.652E+05	719			
<i>Orthodontic patients average waiting times</i>					
Model	1.095E+05	35	3128.17	122.67	< 0.0001
A-Receptionists	1.48	1	1.48	0.0580	0.8098
B-Patient Flow	1.47	1	1.47	0.0575	0.8105
C-Rules	95662.19	2	47831.09	1875.67	< 0.0001
D-Schedules	3830.93	2	1915.47	75.11	< 0.0001
AB	7.88	1	7.88	0.3090	0.5785
AC	106.59	2	53.29	2.09	0.1245
AD	46.38	2	23.19	0.9095	0.4032
BC	45.37	2	22.69	0.8896	0.4113
BD	55.67	2	27.83	1.09	0.3363
CD	9234.71	4	2308.68	90.53	< 0.0001
ABC	31.40	2	15.70	0.6157	0.5405
ABD	65.82	2	32.91	1.29	0.2758
ACD	75.98	4	18.99	0.7449	0.5616
BCD	268.02	4	67.01	2.63	0.0336
ABCD	52.00	4	13.00	0.5098	0.7285
Pure Error	17442.56	684	25.50		
Cor Total	719				

Bold factors and their interactions mean the terms are significant

Rules: Appointment rules, Schedules: Orthodontist working schedules

Sum of squares is type II

df: Degrees of freedom



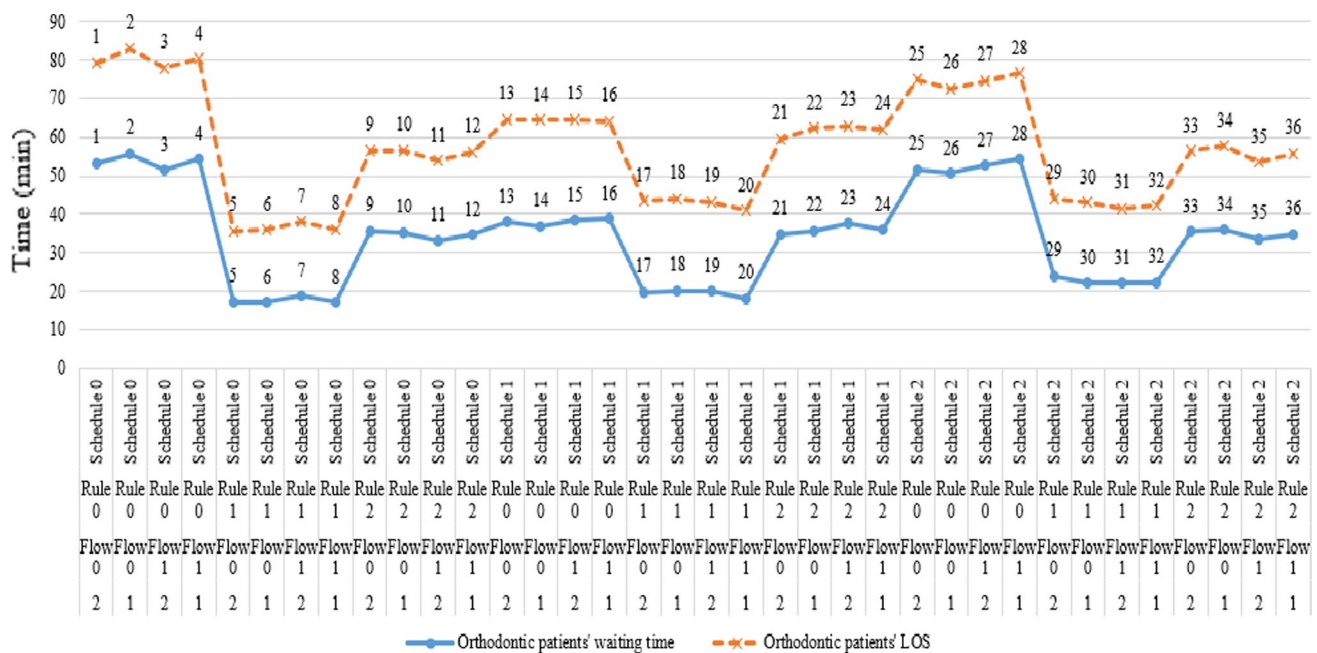


Fig. 3 Average orthodontic patients LOS and waiting times for all experiments

decrease under the same conditions and are highly positively correlated.

As mentioned before, the appointment scheduling rules factor has the greatest impact on both performance measures and this is shown clearly in Fig. 3. Patients average LOS and waiting times range from 35.57 to 83.12 min and from 16.92 to 55.72 min, respectively. Appointment rule 1 has the least average LOS and waiting times in all factors combinations with a range of 35.57 to 44.05 min and 16.92 to 23.88 min, respectively, for experiments 5–8, 17–20 and 29–32. Patients' averages LOS and waiting times increase in appointment rule 2 with a range of 53.4 to 62.87 min and 33.09 to 37.44 min, respectively, for experiments 9–12, 21–24 and 33–36. The highest average LOS and waiting times are for the current rule with a range of 64.14 to 83.12 min and 36.99 to 55.72 min, respectively, for experiments 1–4, 13–16 and 25–28.

On the other hand, the orthodontist working schedules effect on patients average LOS and waiting times differs based on appointment scheduling rules.

For appointment rule 1, orthodontist schedule 0 gives the least patients average LOS and waiting times as in experiments 5, 6, 7, 8 and 20. For appointment rule 2, orthodontist schedule 0 and orthodontist schedule 2 give the least patients average LOS and waiting times as in experiments 11, 12, 35 and 36. For the current rule, orthodontist schedule 1 gives the least patients average LOS and waiting times as in experiments 13, 14, 15 and 16. By applying orthodontist schedule 0 and appointment rule 1, patients average LOS and waiting times can be reduced by 55.03% and 68.19%, respectively.

4.1.2 Results of Orthodontist Utilizations and Number of Orthodontic Patients Served

A full-factorial ANOVA on number of patients served and orthodontist average utilizations is shown in Table 6. Appointment scheduling rules and orthodontist working schedules have significant impact on both performance measures. Orthodontist working schedules have the highest effect on the number of patients served. (Its mean square is 2.735E+06.) In contrast, appointment scheduling rules have the highest effect on orthodontist average utilizations. (Its mean square is 6087.53.) Patient flows and number of receptionists have no effect on both performance measures.

On the other hand, ANOVA shows one significant double interaction for both performance measures which are orthodontist working schedules and appointment scheduling rules. Also, an interaction between all decision factors is shown to be nearly significant.

Figures 4 and 5 represent the number of patients served and orthodontist average utilizations for all experiments.

As mentioned before, the orthodontist working schedule factor has the greatest effect on the number of patients served which increase dramatically when orthodontist working schedules change. The number of patients served range from 397.75 to 679 patients during the experiments. Orthodontist working schedule 2 has the highest number of patients served with a range of 613.5 to 679 patients for experiments 25 to 36. Orthodontist working schedule 1 has a lower number of patients served than schedule 2 with a range of 586 to 588 patients for experiments 13–24. Orthodontist



Table 6 ANOVA on orthodontist average utilizations and number of patients served

Source	Sum of squares	df	Mean square	F-value	P-value
<i>Number of patients served</i>					
Model	6.178E+06	35	1.765E+05	730.37	< 0.0001
A-Receptionists	726.01	1	726.01	3.00	0.0835
B-Patient Flow	27.61	1	27.61	0.1142	0.7355
C-Rules	6.793E+05	2	3.397E+05	1405.30	< 0.0001
D-Schedules	5.469E+06	2	2.735E+06	11314.72	< 0.0001
AB	161.50	1	161.50	0.6682	0.4140
AC	875.48	2	437.74	1.81	0.1643
AD	163.86	2	81.93	0.3390	0.7126
BC	195.91	2	97.95	0.4053	0.6669
BD	780.61	2	390.30	1.61	0.1997
CD	23192.49	4	5798.12	23.99	< 0.0001
ABC	497.55	2	248.78	1.03	0.3578
ABD	294.64	2	147.32	0.6095	0.5439
ACD	870.14	4	217.54	0.9000	0.4634
BCD	791.51	4	197.88	0.8187	0.5134
ABCD	1111.95	4	277.99	1.15	0.3318
Pure Error	1.653E+05	684	241.69		
Cor Total	6.344E+06	719			
<i>Orthodontist average utilizations</i>					
Model	16183.96	35	462.40	97.90	< 0.0001
A-Receptionists	1.18	1	1.18	0.2491	0.6179
B-Patient Flow	3.09	1	3.09	0.6545	0.4188
C-Rules	12175.07	2	6087.53	1288.85	< 0.0001
D-Schedules	3159.89	2	1579.94	334.51	< 0.0001
AB	0.3336	1	0.3336	0.0706	0.7905
AC	2.01	2	1.00	0.2123	0.8088
AD	3.71	2	1.85	0.3926	0.6755
BC	7.17	2	3.59	0.7590	0.4685
BD	5.58	2	2.79	0.5906	0.5543
CD	712.43	4	178.11	37.71	< 0.0001
ABC	5.67	2	2.84	0.6007	0.5487
ABD	8.63	2	4.32	0.9140	0.4014
ACD	15.03	4	3.76	0.7955	0.5283
BCD	39.15	4	9.79	2.07	0.0828
ABCD	45.03	4	11.26	2.38	0.0501
Pure Error	3230.68	684	4.72		
Cor Total	19414.65	719			

Bold factors and their interactions mean the terms are significant

Rules: Appointment rules, Schedules: Orthodontist working schedules

Sum of squares is type II

df: Degrees of freedom

working schedule 0 which is the current schedule used in the clinic has the lowest number of patients served with a range of 397 to 472 patients for experiments 1–12. Also, orthodontist working schedules have a clear effect on the number of patients served with different appointment scheduling rules. Appointment rule 2 results in highest number of patients served with all orthodontist schedules as in experiments 33–36, 21–24 and 9–12 for schedules 2, 1 and 0, respectively, followed by appointment rule 1 in experiments 29–32, 17–20 and 5–8 for schedules 2, 1 and 0, respectively. Appointment rule 0 has the lowest number of patients served with all schedules as in experiments 25–28, 13–16 and 1–4 for schedules 2, 1 and 0, respectively. By applying appointment rule 2 with orthodontist schedule 2, the number of patients served can be increased by 70.7%. As shown in Fig. 5, almost all orthodontist average utilizations are high in all experiments. The appointment scheduling rules factor has the greatest effect on orthodontist average utilizations. As in number of patients served, the highest orthodontist average utilizations are observed with appointment scheduling rule 2 which range from 83.68 % to 91.11%, followed by appointment rule 1 which range from 76.7 % to 82.18 % and appointment rule 0 which has the lowest orthodontist average utilizations with a range of 74.11% to 80.39%. In addition, orthodontist schedules 0 and 2 have high orthodontist average utilizations with all appointment scheduling rules compared to orthodontist schedule 1. By applying orthodontist schedule 0 and appointment rule 2, orthodontist average utilizations can be increased by 15.73%.

4.1.3 Results of Primary Diagnosis Patients LOS

Table 7 shows ANOVA for primary diagnosis patients average LOS; as shown patient flow is the only factor that has a significant effect with a mean square of 4149.75. In addition, there is only one significant interaction effect between the number of receptionists, patient flows and orthodontist working schedules. Since orthodontist working schedules affect significantly the number of orthodontic patients and increase them up to 70.7%, this interaction can be explained because all patients from different clinics share the same reception, accounting and radiographies, and by increasing the number of patients in the clinics, crowding will also increase in shared services and therefore the interaction of number of receptionists, orthodontist working schedules and patient flows can affect primary diagnosis patients average LOS.

Figure 6 shows the average LOS of primary diagnosis patients for all experiments; the figure shows a cyclical pattern in which all experiments with patient flow 1 reduce primary diagnosis patients average LOS. Patients average LOS ranges from 49.78 to 59.68 min. Having two receptionists and applying patient flow 1, orthodontists working schedule



Table 7 ANOVA of primary diagnosis patients average LOS.

Source	Sum of squares	df	Mean square	F-value	P-value
Model	5228.22	35	149.38	4.65	< 0.0001
A-Receptionists	68.58	1	68.58	2.13	0.1446
B-Patient flow	4149.75	1	4149.75	129.05	< 0.0001
C-Rules	56.73	2	28.36	0.8820	0.4144
D-Schedules	99.04	2	49.52	1.54	0.2151
AB	24.27	1	24.27	0.7548	0.3853
AC	2.89	2	1.44	0.0449	0.9561
AD	131.39	2	65.70	2.04	0.1304
BC	21.81	2	10.91	0.3391	0.7125
BD	37.53	2	18.77	0.5836	0.5582
CD	65.35	4	16.34	0.5081	0.7298
ABC	37.32	2	18.66	0.5803	0.5600
ABD	201.92	2	100.96	3.14	0.0439
ACD	103.76	4	25.94	0.8067	0.5211
BCD	202.97	4	50.74	1.58	0.1784
ABCD	24.90	4	6.23	0.1936	0.9417
Pure error	21995.53	684	32.16		
Cor total	27223.75	719			

Bold factors and their interactions mean the terms are significant

Rules: Appointment rules, Schedules: Orthodontist work schedules

Sum of squares is type II

df: Degrees of freedom

receptionists from 25.5 to 30.5%, and it clearly shows how removing one receptionist affects average utilizations. By removing one receptionist and applying patient flow 1, appointment rule 2 and orthodontist schedule 2,

the receptionist average utilizations can be increased by 65.82%.

4.2 Comparison of Traditional and Modified TOPSIS Results

It is infeasible to find one experiment that absolutely optimizes all performance measures simultaneously. In this research, TOPSIS is used to rank the suggested experiments based on performance measures collected from each experiment. TOPSIS provides a final score for each experiment; the experiment with the highest score improves performance measures more than others and is considered the best alternative.

Each experiment is input as an alternative in both modified and traditional TOPSIS techniques, and each performance measure appears as a criterion. Table 9 shows TOPSIS decision matrix which includes the alternatives (experiments) and their corresponding criteria (performance measures), w_j represents relative weights of criteria, and in this research, all criteria weights are assumed equal. Orthodontic patients LOS and waiting times and primary diagnosis patients LOS are cost criteria and number of orthodontic patients served, and utilizations of orthodontists and receptionists are benefit criteria. Modified and traditional TOPSIS techniques are implemented using MATLAB R2017b.

In this research, some of the selected criteria (performance measures) are correlated. Table 10 shows the correlation coefficients of the criteria.

As shown in bold, the correlation between the orthodontic patients average LOS (C1) and waiting times (C2) criteria is a very strong positive relationship; when LOS decreases, waiting times also decrease and vice versa. Also, correlation between the number of orthodontic patients served

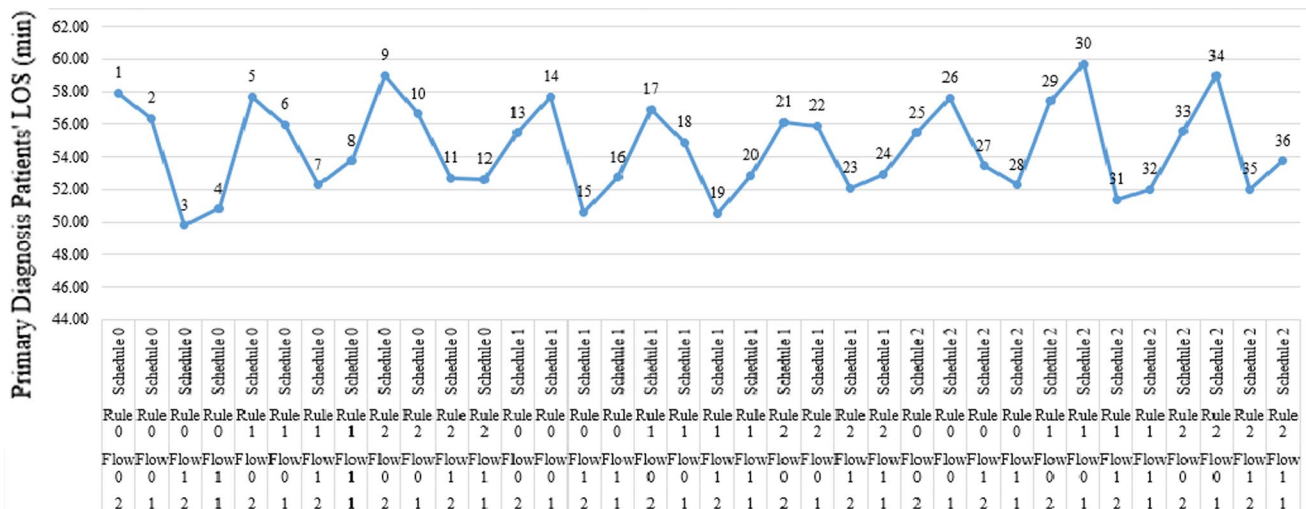
**Fig. 6** Primary diagnosis patients' average LOS for all experiments

Table 8 ANOVA on receptionist average utilizations

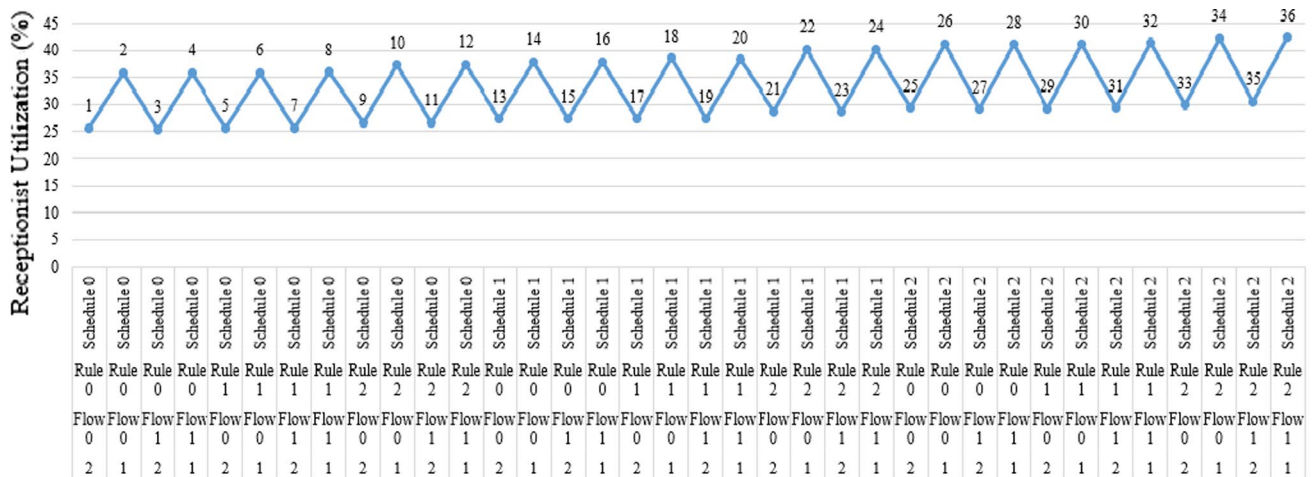
Source	Sum of squares	df	Mean square	F-value	P-value
Model	25247.65	35	721.36	936.91	< 0.0001
A-Receptionists	22521.53	1	22521.53	29251.10	< 0.0001
B-Patient Flow	0.2537	1	0.2537	0.3295	0.5662
C-Rules	288.11	2	144.06	187.10	< 0.0001
D-Schedules	2327.13	2	1163.57	1511.25	< 0.0001
AB	0.0293	1	0.0293	0.0381	0.8453
AC	9.06	2	4.53	5.88	0.0029
AD	81.17	2	40.58	52.71	< 0.0001
BC	1.34	2	0.6718	0.8725	0.4184
BD	1.86	2	0.9295	1.21	0.2997
CD	9.73	4	2.43	3.16	0.0138
ABC	0.2792	2	0.1396	0.1813	0.8342
ABD	0.6205	2	0.3102	0.4029	0.6685
ACD	2.62	4	0.6556	0.8515	0.4928
BCD	2.54	4	0.6338	0.8232	0.5106
ABCD	1.38	4	0.3446	0.4475	0.7742
Pure Error	526.64	684	0.7699		
Cor Total	25774.29	719			

Bold factors and their interactions mean the terms are significant

Rules: Appointment rules, Schedules: Orthodontist working schedules

Sum of squares is type II

df: Degrees of freedom

**Fig. 7** Receptionist average utilizations for all experiments

(C3) with both the orthodontist average utilizations (C4) and the receptionist average utilizations (C6) is a moderately positive relationship; when the number of patients served increases, receptionist and orthodontist utilizations increase.

The correlations between criteria cannot be neglected which explains why introducing the Mahalanobis distance that takes them into consideration instead of the Euclidean distance in traditional TOPSIS is important and can help in improving the results.

Table 11 shows the resulting scores, ranks and differences in both traditional and modified TOPSIS techniques for each alternative. The table clearly shows how the consideration of correlation between criteria affects the results. In modified TOPSIS, Alternative 36 has the highest score and therefore it has the first rank among all alternatives, in contrast to traditional, Alternative 36 has a lower score and is ranked 13th with a difference of 12 ranks between modified and traditional TOPSIS



Table 9 TOPSIS decision matrix for the current research

A_i	C1	C2	C3	C4	C5	C6	A_i	C1	C2	C3	C4	C5	C6
1	79.1	53.2	397.8	78.8	57.9	25.7	20	40.9	18.0	522.1	77.6	52.8	38.3
2	83.1	55.7	397.9	77.9	56.3	35.9	21	59.6	34.8	588.3	83.9	56.1	28.7
3	78.0	51.4	400.1	77.3	49.8	25.5	22	62.3	35.5	585.2	83.7	55.9	40.2
4	80.4	54.3	399.6	77.6	50.8	35.9	23	62.9	37.4	587.0	84.7	52.1	28.6
5	35.6	17.2	415.9	81.1	57.7	25.7	24	61.9	35.8	579.3	84.4	52.9	40.3
6	35.9	16.9	416.3	81.2	55.9	35.9	25	74.8	51.6	618.4	79.9	55.5	29.4
7	37.9	18.7	415.4	82.2	52.3	25.8	26	72.5	50.5	618.4	80.4	57.6	41.2
8	36.1	16.9	418.7	82.0	53.8	36.0	27	74.7	52.6	613.5	79.9	53.4	29.2
9	56.7	35.7	462.2	90.7	58.9	26.8	28	76.6	54.3	615.3	80.3	52.3	41.1
10	56.5	35.1	461.8	90.6	56.6	37.3	29	44.0	23.9	628.9	81.6	57.4	29.2
11	53.9	33.1	472.1	91.1	52.7	26.7	30	42.9	21.9	625.6	81.4	59.7	41.1
12	56.3	34.6	465.1	90.3	52.6	37.4	31	41.4	22.0	632.8	81.7	51.4	29.3
13	64.7	37.9	492.4	74.1	55.5	27.3	32	42.1	22.1	628.3	81.4	52.0	41.4
14	64.3	36.9	495.8	74.9	57.7	37.9	33	56.4	35.4	679.1	87.4	55.5	29.9
15	64.7	38.4	499.7	76.0	50.6	27.4	34	57.6	35.8	672.3	86.9	58.9	42.4
16	64.1	38.7	486.0	74.1	52.7	37.9	35	53.4	33.4	678.3	86.6	52.0	30.5
17	43.3	19.8	519.5	77.3	56.9	27.4	36	55.8	34.7	673.8	87.6	53.7	42.6
18	44.1	19.9	519.7	77.4	54.9	38.6	w_j	1	1	1	1	1	1
19	43.1	19.9	515.8	76.7	50.6	27.3							

A_i : Alternatives (Experiments), w_j : Weight of each criteria

C1: Orthodontic patients average LOS, C2: Orthodontic patients average waiting times

C3: Number of orthodontic patients served, C4: Orthodontist average utilizations

C5: Primary diagnosis patients average LOS, C6: Receptionist average utilizations

Table 10 Correlation coefficients of the criteria

	C1	C2	C3	C4	C5	C6
C1	1	0.986971	− 0.04897	− 0.17402	− 0.06333	0.017897
C2	−	1	− 0.00883	− 0.08972	− 0.05061	0.016874
C3	−	−	1	0.254587	0.070168	0.311186
C4	−	−	−	1	0.148337	0.07517
C5	−	−	−	−	1	0.137119
C6	−	−	−	−	−	1

C1: Orthodontic patients average LOS, C2: Orthodontic patients average waiting times

C3: Number of orthodontic patients served, C4: Orthodontist average utilizations

C5: Primary diagnosis patients' average LOS, C6: Receptionist average utilizations

techniques. In traditional TOPSIS, Alternative 32 has the first rank and highest score and is ranked second in modified with a difference of one rank which clearly shows that this alternative is robust. Alternative 1 has the same rank which is 36th in both techniques, which shows that the performance measures of the current system are the worst among all alternatives and changes must be applied to improve the current system.

Alternatives 32 and 36 which were obtained from traditional and modified TOPSIS, respectively, improved the EDC as follows:

- Reduced orthodontic patients' averages LOS and waiting times by 46.74% and 58.46%, respectively, for Alternative 32 and by 29.44% and 37.74% for Alternative 36.
- Increased number of orthodontic patients served by 57.95% for Alternative 32 and by 69.39% for Alternative 36.
- Increased orthodontist average utilizations by 3.39% for Alternative 32 and by 11.22% for Alternative 36.
- Reduced primary diagnosis patients' average LOS by 10.19% for Alternative 32 and by 7.12 for Alternative 36.



Table 11 Comparison of results between modified and traditional TOPSIS techniques

Alternative	Modified TOPSIS		Traditional TOPSIS		Difference
	Resulting score	Rank	Resulting score	Rank	
1	0.170525	36	0.074751	36	0
2	0.31463	35	0.177892	34	1
3	0.435329	27	0.144668	35	8
4	0.468287	23	0.203132	33	10
5	0.411228	30	0.648556	11	19
6	0.504755	19	0.714336	7	12
7	0.529721	16	0.644535	12	4
8	0.565615	9	0.719681	6	3
9	0.421768	29	0.455169	25	4
10	0.539088	12	0.524868	20	8
11	0.575203	8	0.508411	21	13
12	0.640327	3	0.53883	19	16
13	0.320894	34	0.385049	28	6
14	0.366212	33	0.460823	23	10
15	0.476396	21	0.392617	27	6
16	0.469485	22	0.443674	26	4
17	0.407165	31	0.671041	10	21
18	0.528256	17	0.75745	4	13
19	0.548487	11	0.675679	9	2
20	0.588625	7	0.781945	3	4
21	0.458746	24	0.503679	22	2
22	0.536534	14	0.546241	18	4
23	0.558331	10	0.460057	24	14
24	0.618823	5	0.547755	17	12
25	0.385938	32	0.275547	31	1
26	0.430141	28	0.368508	29	1
27	0.438858	26	0.269618	32	6
28	0.533248	15	0.338792	30	15
29	0.456309	25	0.701344	8	17
30	0.485049	20	0.811845	2	18
31	0.612501	6	0.736318	5	1
32	0.689833	2	0.845997	1	1
33	0.537857	13	0.555562	16	3
34	0.526601	18	0.603813	14	4
35	0.632365	4	0.600054	15	11
36	0.702924	1	0.635112	13	12

- Increased receptionist average utilizations by 61.01% for Alternative 32 and by 65.83% for Alternative 36.

Figure 8 represents the resulting scores plot of all alternatives in both modified and traditional TOPSIS techniques. Traditional TOPSIS does not take correlations between criteria into consideration which amplifies the resulting scores of alternatives with the best criteria

(performance measures) values and also amplifies differences between alternatives as confirmed by [20].

In contrast, modified TOPSIS takes criteria correlations into consideration and therefore provides more accurate results. Resulting scores of alternatives in modified TOPSIS are more consistent and have lower variability. Table 12 shows mean and standard deviation of resulting scores for modified and traditional TOPSIS techniques. The table confirms that the modified TOPSIS has lower variability (its standard deviation value is 0.109528) than traditional one. (Its standard deviation value is 0.197802.)

5 Conclusions and Future Work

This paper focused on solving the most important issues in EDS by integrating DES and DOE with MCDM analysis. DOE was used to study the effect of suggested decision factors and their interactions on selected performance measures to help improve the system. Several decision factors were studied such as appointment scheduling rules for orthodontic patients, orthodontists working schedules, patient flows and number of receptionists on several performance measures such as orthodontic and primary diagnosis patients LOS, orthodontic patients waiting times, number of patients served and orthodontist and receptionist utilizations. Several scenarios representing all possible combinations of factors levels were applied in simulation models. The selected performance measures were collected from each scenario to study how the decision factors and their interactions affected performance measures. Scenarios were then input to traditional and modified TOPSIS techniques as alternatives to be compared and ranked based on multiple criteria represented by the performance measures computed by DES models with the goal of selecting the best alternative. The advantage of modified TOPSIS is that it takes into consideration correlations between criteria (performance measures). The following points summarize the conclusions of the study and provides suggestions for future work.

- Orthodontic patients average LOS and waiting times had the same behavior during the experiments; both are affected significantly by appointment scheduling rules, orthodontists working schedules and their interaction.
- The number of orthodontic patients served is proportionally related to orthodontist average utilizations and is both affected significantly by appointment scheduling rules, orthodontist working schedules and their interaction.
- The effect of patients flows factor is not significant on orthodontic patients average LOS due to the low percentage of patients that benefit from the change in patient flows. In contrast, 77 % of primary diagnosis patients that



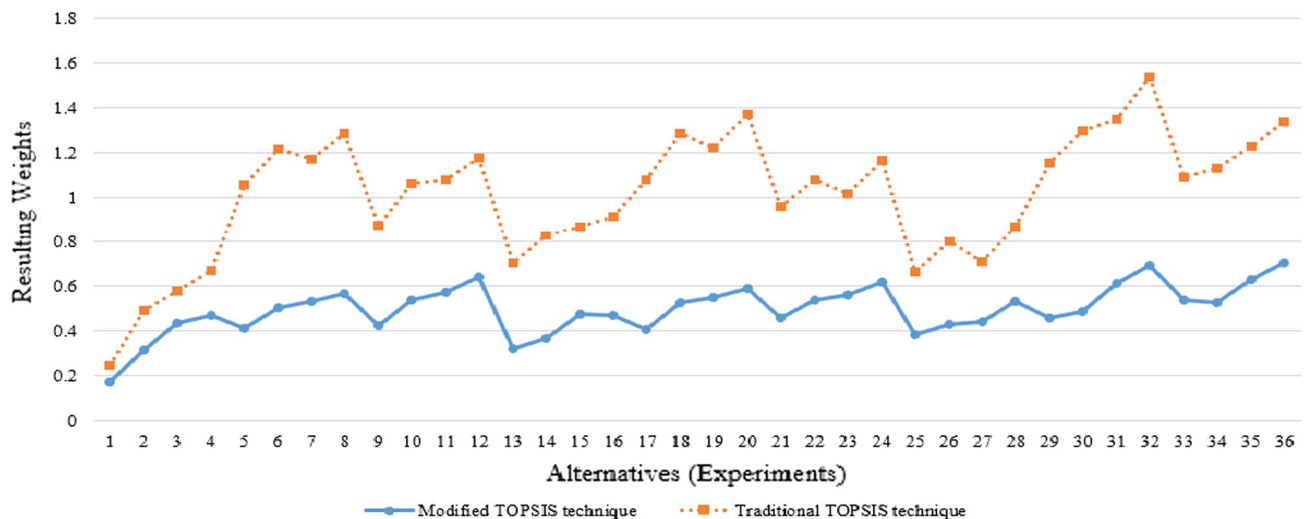


Fig. 8 The comparison of scores obtained from modified and traditional TOPSIS technique

Table 12 Mean and standard deviation of modified and traditional TOPSIS techniques

Technique	Mean	Standard deviation
Modified TOPSIS	0.496834	0.109528
Traditional TOPSIS	0.520093	0.197802

used the new patient flows benefitted from the change and this had a significant effect on primary diagnosis patients average LOS.

- The number of receptionists, appointment scheduling rules and orthodontist working schedules had a significant effect on receptionist average utilizations.
- Each performance measure is affected by the decision factors and their levels differently, the factor level that is appropriate for a specific performance measures is not necessarily suitable for another, this means that it is infeasible to find an experiment that optimizes all performance measures simultaneously, and therefore, MCDM analyses such as the TOPSIS can help in ranking experiments to obtain the best combination of factors levels.
- Modified and traditional TOPSIS techniques provide different results due to correlations among the criteria (performance measures).
- Traditional TOPSIS amplifies the resulting scores of alternatives that have better criteria values because it cannot deal with correlations between them. In contrast, the resulting scores from modified TOPSIS are less variable and more robust as compared to traditional TOPSIS.

Future work suggestions are summarized in the following points:

- This research does not take into consideration orthodontists lateness and absence, and also, it assumes that orthodontic patients arrive to the clinic exactly at their appointments times. As a future work, orthodontists' and patients' lateness and absence can be considered to study the effect of these behaviors on the clinic's performance.
- In this work, two appointment scheduling rules were studied in the orthodontic clinic which were individual block/fixed interval (one patient assigned an appointment every 15 minutes) and multiple block/fixed interval (three patients assigned an appointment every 30 minutes). There are many other appointment rules that can be tested and compared in future projects to obtain the best results.
- The same work procedure can be applied to other dentistry clinics that are already constructed in the simulation models such as periodontics, restorative, surgery, kids and prosthodontics. Also, similar healthcare systems can apply the same procedure to improve their performance.
- Other MCDM techniques that take criteria correlations into consideration can be used to rank the experiments other than modified TOPSIS such as the analytic network process (ANP).
- The weights assigned to the evaluation criteria can be obtained from field experts and utilized in the proposed method.

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